

Cyber Threat Detection System

Real-Time Threat Detection in a Data Diode Environment — Voice-Over Script

TOTAL RUNTIME	TARGET AUDIENCE	STONE & STYLE	VIDEO FILE
90 Seconds (1:30)	Academic Reviewers / Judges	Technical, Confident, SOC Lead	CyberThreat_Detection_Demo.mp4

1. Introduction & Data Diode Architecture

0:00 – 0:18

On-Screen Action: The SOC Dashboard loads on the Live Monitor tab. Real-time telemetry graphs begin ticking. Status chips show "Read-Only Unidirectional" and "Live Backend".

VOICE NARRATION:

"Welcome everyone. Today, I am demonstrating our **Real-Time AI Cyber Threat Detection System**, specially designed for **Data Diode and Unidirectional Network Environments**. In high-security defense and industrial facilities, data diodes enforce a strict physical, one-way fiber path to guarantee zero return traffic. Our system ingests and correlates this unidirectional packet stream in real time, detecting sophisticated threats without requiring network handshakes."

2. Live Telemetry & Attack Simulation

0:18 – 0:38

On-Screen Action: Pointer moves to the Attack Simulation panel. Clicking **SYN Flood**. Live throughput spikes immediately and a high-priority alert pops into the Active Threats feed.

VOICE NARRATION:

"On the main dashboard, real-time WebSocket telemetry tracks packets per second, throughput bandwidth, and active threat counts. Now, let's simulate an active volumetric attack by triggering a **SYN Flood**... [Pause 1.5s for alert to populate] Notice how our streaming sliding-window engine immediately identifies the abnormal TCP handshake ratio, raising a Critical alert within milliseconds."

3. Deep Threat Inspection & AI Incident Mitigation

0:38 – 0:58

On-Screen Action: Clicking the newly generated alert row. The Deep Threat Inspection Modal slides into view, showcasing feature heuristics, confidence scores, and recommended actions.

VOICE NARRATION:

"Clicking on the alert reveals the **Deep Threat Inspection Modal**. SOC analysts get an instantaneous breakdown of the attack vector: flow telemetry, rule heuristics, and our machine-learning confidence score. The engine correlates packet size entropy, connection frequency, and DNS n-gram anomalies. Crucially, it provides automated, actionable mitigation commands—such as immediate firewall drop rules and internal host quarantine—allowing analysts to contain threats in seconds."

4. Topology Lab & Physical Boundary Mapping

0:58 – 1:16

On-Screen Action: Clicking the **Topology** tab. An interactive network graph visualizes traffic crossing from the external network through the optical diode into the secure enclave.

VOICE NARRATION:

"Next, let's navigate to the **Topology Lab**. This interactive visualizer maps our physical isolation boundary. You can see untrusted boundary interfaces feeding traffic across the hardware diode into our secure enclave. The engine dynamically monitors the boundary to verify that reverse return paths remain physically blocked, ensuring zero risk of Command-and-Control reaching back outside."

5. Forensic Scanner & Concluding Summary

1:16 – 1:30

On-Screen Action: Switching to the **Forensics** tab showing packet replay tools, then returning to the **Live Monitor** with continuous streaming charts.

VOICE NARRATION:

"Finally, the **Forensic Scanner** allows deep packet inspection and replay of real-world PCAP files and Zeek logs for thorough incident audits. By combining hardware-enforced unidirectional safety with intelligent streaming ML, this platform ensures proactive threat defense for mission-critical enclaves. Thank you, and I am happy to answer any questions."

Anticipated Question	Model Response / Speaking Point
Why is threat detection harder on a Data Diode?	Because data diodes are physically unidirectional, standard TCP bidirectional ACKs and handshakes cannot complete back to the sender. Threat detection must rely strictly on continuous, single-direction flow heuristics and sliding-window feature analysis.
What ML models are powering detection?	A hybrid ensemble: Random Forest for multi-class flow classification, character n-gram entropy scoring for DNS covert exfiltration, and adaptive threshold correlation across sliding time windows.
How are false positives prevented?	Through multi-stage correlation: low-level packet heuristics must align with sliding-window connection frequency thresholds before an alert escalation is triggered.

Quick PowerPoint Integration Checklist

- **Video File:** Located at C:\Users\vivek\OneDrive\Desktop\CyberThreat_Detection_Demo.mp4
- **Slide Insertion:** Insert → Video → This Device → Choose the MP4 file.
- **Playback Configuration:** Under the Playback tab, select *Start: Automatically* and check *Loop until Stopped*.
- **Audio Narration:** Use Insert → Audio → Record Audio to record your voice reading this script directly into the slide!